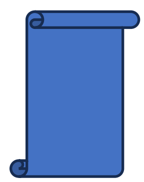
EARTHQUAKE ANALYSIS

PROJECT OVERVIEW

- This project analyzes telecom customer churn to understand the factors influencing customer retention, revenue generation, refunds, and overall profitability. With **27% of customers churned, 6% newly joined, and 67% retained**, the analysis focuses on identifying behavioral and demographic patterns associated with different customer outcomes.
- The analysis examines **customer revenue, refunds, tenure, service charges, contract types, status and demographic characteristics** to identify high-value customers, understand churn patterns and uncover relationships between customer attributes and financial performance. Using **univariate, bivariate and multivariate analysis**, the project establishes a descriptive view of the customer base while developing business-oriented mappings of potential drivers behind unwanted outcomes.



THE PROBLEM STATEMENT:

- The telecom business is experiencing a significant customer churn rate, with **27% of customers leaving the service**, creating potential revenue loss and increasing the need for effective retention strategies. However, not all customers contribute equally to revenue or profitability, making it important to understand **which customer segments are most valuable and which characteristics are associated with churn or unfavorable financial outcomes**.
- The key business challenge is to determine how **tenure, revenue, refunds, service charges, contract types, customer status, and demographic attributes** influence customer behavior and profitability. The analysis aims to identify patterns among churned, retained and newly joined customers and determine whether high-revenue customers also demonstrate lower refund activity and stronger retention.
- By addressing these questions, the project seeks to provide actionable insights that can help the business **identify high-value customers, recognize potential churn patterns, improve retention efforts, reduce revenue leakage and strengthen customer lifetime value**.

🔹 📁 DATASET DETAILS

- The analysis uses the **Telecom Customer Churn dataset from California for Q2 2022, sourced from Maven Analytics.** The dataset contains **7,044 customer records across 38 columns**, providing a comprehensive view of customer demographics, subscription behavior, billing activity, revenue and churn status.
- Key attributes include **Customer ID, Age, Gender, Tenure, Contract Type, Monthly Charges, Total Charges, Total Refunds, Extra Data Charges, Long-Distance Charges, Total Revenue and Customer Status (Churned, Stayed, or Joined).** Geographic information, including ZIP Code and City for location-based analysis.
- As part of the data **preparation process, data types, missing values, duplicate records and potential data-quality issues** were reviewed and addressed before visualization. Certain numeric fields contain zero values where services were not subscribed to. For example, long-distance charges for customers without phone service. Billing-related fields were also assessed for missing or duplicate values to ensure the dataset was appropriately prepared for analysis in Tableau.
- **Maven Analytics**: https://mavenanalytics.io/data-playground/telecom-customer-churn

🔹 WHAT'S INSIDE

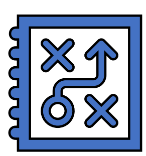
- 👥 **Customer Churn Overview** : The project begins with an overview of the customer base, highlighting the distribution **of Churned (27%), Stayed (67%), and Joined (6%) customers**. This establishes the business context and provides an initial understanding of the scale of customer retention and churn.
- 💰 **Revenue & Refund Analysis** : Revenue and refund activity are examined to understand the financial contribution and potential dissatisfaction signals within the customer base. Interactive aggregation allows users to evaluate **Sum, Average, Minimum, and Maximum values, providing multiple perspectives on customer financial behavior.**
- 📈 **Tenure & Spending Behavior**: The analysis explores the relationship between customer tenure and spending, examining how monthly charges and total charges evolve as customers

remain with the company. This section helps demonstrate the importance of long-term customer relationships in driving revenue growth.

- 🌐 **Service Usage & Customer Behavior :** Multiple service-related metrics are analyzed against tenure, including **Extra Data Charges and Long-Distance Charges**. This provides insight into how service consumption develops throughout the customer lifecycle and whether longer-tenure customers expand their usage across different service categories.
- 🔎 **High-Value Customer Identification** : The project includes a dedicated analysis of high-value customers using revenue and refund rankings. **A dynamic Top 10 parameter** and ranking logic help identify customers who generate significant revenue while demonstrating comparatively lower refund activity.
- 🌎 **Geographic & Demographic Context** : The dashboard incorporates external geographic context through Google/Wikipedia integration, allowing users to select a city and explore additional demographic information. This connects customer-level analysis with broader geographic characteristics.
- 📖 **Business Storytelling** : Rather than presenting disconnected visualizations, the project follows a structured analytical narrative:

Customer Landscape → Revenue & Refunds → Tenure → Spending → Service Usage → Customer Profiles → High-Value Customers → Retention Opportunities. This progression helps translate individual analytical findings into a unified business story.

- 🔮 **Future Analytical Opportunities** : The project also establishes a foundation for future **predictive analysis, including churn prediction modeling, customer lifetime value modeling, deeper demographic segmentation, and proactive identification of at-risk customers**. The project combines business storytelling to answer a central business question: **Which customers create the greatest long-term value, what behaviors influence that value, and how can the business retain and nurture them?**



DAX FORMULAS & NEW MEASURES

- The project used Tableau's native analytical capabilities to dynamically evaluate customer and financial metrics. A parameter named **"Revenue & Refund Aggregation"** was created to allow users to dynamically switch between **Sum, Average, Minimum, and Maximum** values for Revenue and Refund metrics.

- The aggregation parameter enabled the same visualization to answer multiple analytical questions without requiring separate worksheets for each calculation. For example, users could examine:
  - **Sum:** Overall revenue or refund contribution.
  - **Average:** Typical revenue or refund value per customer.
  - **Minimum:** Lowest observed customer-level value.
  - **Maximum:** Highest observed customer-level value.
- Additional analytical calculations and controls included **Top 10 Limit**, revenue ranking in descending order, refund ranking in ascending order and filtering logic to identify customers who combine high revenue contribution with relatively low refund activity.
- Average calculations were also incorporated into comparative visualizations, including **Average Monthly Charge by Tenure** and **Average Total Charges by Customer Status**. **Average lines and trend lines were** used to provide analytical benchmarks and make differences between customer groups easier to interpret.

DATA MODELING

- The analytical structure was designed to support three levels of analysis:
  - **Customer-level analysis:** Examining individual customers based on revenue, refunds, tenure, demographic attributes, service usage, and customer status.
  - **Segment-level analysis:** Comparing customer groups based on status, contract type, tenure, and other relevant characteristics.
  - **Metric-level analysis:** Evaluating financial and service-related measures such as Total Revenue, Refunds, Monthly Charges, Total Charges, Extra Data Charges and Long-Distance Charges. Interactive filtering by **tenure, contract type and customer status** further supported cohort-level exploration and allowed relationships to be examined within specific customer groups. The data structure also supported the identification of top-performing customers through revenue and refund ranking, enabling the analysis to focus on customers who contribute higher revenue while maintaining relatively low refund activity.

VISUALS UTILIZED TO ANALYZE THE DATA

1. Dynamic KPI Cards - Revenue & Refund Analysis : A dynamic KPI card was created using the **“Revenue & Refund Aggregation” parameter.** Users can switch between **Sum, Average, Minimum, and Maximum** values to evaluate the distribution and scale of revenue and refund activity.

2. Line Chart - Average Monthly Charge by Tenure : A line chart was used to analyze the relationship between customer **tenure and Average Monthly Charge**. The visualization helps identify how customer spending behavior changes as customers remain with the telecom provider for longer periods. The trend provides an important perspective on **whether longer-tenured customers demonstrate different spending patterns compared with newer customers.**

3. Stacked Bar Chart - Average Total Charges by Customer Status: A stacked bar chart was developed to compare Average Total Charges **across Joined, Stayed, and Churned customers**. An Average Line was incorporated to provide a benchmark against which each customer-status group could be compared.

4. Dual-Trend Visualization - Service Charges by Tenure : A dual-trend visualization was developed to examine multiple service-related measures simultaneously across tenure. The visualization combines Average **Extra Data Charges using a line chart with circle marks, Average Long-Distance Charges using an area chart,** Tenure as the common analytical dimension and Trend lines to highlight underlying patterns.

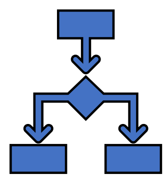
5. Detailed Customer-Level Table: The table includes: **Customer ID, Revenue, Refunds, Average GB Download, Age, Gender, Customer Status, Internet Type and Unlimited Data**. The table was particularly useful for identifying and examining **the characteristics of high-revenue customers. By combining financial, demographic, behavioral and service attributes,** users can move from identifying a high-value customer to understanding what characteristics define that customer.

6. Top 10 Customer Analysis : A **Top 10 Limit parameter** was implemented to control the number of customers displayed in the high-value analysis. **Revenue was ranked in descending order, while refunds were ranked in ascending order.** This approach focuses attention on customers who demonstrate a desirable combination of high revenue contribution and relatively low refund activity, supporting the business objective of identifying customers with stronger profitability potential.

7. Interactive Filters : Interactive dashboard filters were incorporated to allow users to analyze customer behavior across specific cohorts. Users can filter the analysis by Tenure, Contract Type, and Customer Status to determine whether observed relationships remain consistent within different customer segments.

❖ **Overall Analytical Design**

Together, these visualizations create a progression from high-level financial analysis → customer relationship analysis → multivariate behavioral analysis → high-value customer identification. The dashboard design emphasizes interactivity and exploration, allowing stakeholders to investigate not only what is happening, but also which customer segments are contributing to the outcome and what characteristics are associated with those outcomes.



STEPS I FOLLOWED TO BUILD THE DASHBOARD

- The dashboard development followed a structured analytical workflow designed to move from **basic customer understanding to deeper behavioral and business insights**.

1. Defined the Business Objective : I first established the objective of understanding customer churn, revenue contribution, refund activity, tenure, service charges, contract types and demographic characteristics. The analysis was designed to identify **high-value customers, potential churn patterns, and factors associated with customer profitability**.

2. Built the Univariate Analysis : I began with individual-variable analysis to establish a descriptive baseline of the customer dataset. A **Revenue & Refund Aggregation parameter** was created to dynamically switch between **Sum, Average, Minimum, and Maximum** values. KPI cards were used to make these measures easily interpretable and interactive.

3. Developed the Bivariate Analysis : After establishing the baseline, I analyzed relationships between two variables to identify behavioral patterns.

- **Average Monthly Charge by Tenure:** A line chart was used to examine how customer spending changes as tenure increases.
- **Average Total Charges by Customer Status:** A stacked bar chart compared spending behavior across **Joined, Stayed, and Churned** customers.
- **Average Lines:** Average lines were incorporated to provide a benchmark for comparing customer groups and identifying values above or below the overall average.

4. Developed the Multivariate Analysis : A dual-trend visualization compared **Average Extra Data Charges** and **Average Long-Distance Charges** against Tenure, allowing multiple service-related metrics to be evaluated simultaneously. I also created a detailed customer-level table containing **Customer ID, Revenue, Refunds, Average GB Download, Age, Gender, Customer Status, Internet Type, and Unlimited Data**. This provided a more comprehensive view of the characteristics associated with the project's highest revenue contributors.

5. Created the Top Customer Analysis : To identify customers who contribute significant revenue while maintaining lower refund activity, I developed A **Top 10 Limit parameter**, Revenue ranking in **descending order**, Refund ranking in **ascending order** and A Top 10 filter. This approach helped prioritize customers based on both **financial contribution and refund behavior**, rather than evaluating revenue in isolation.

6. Added External Context Through Dashboard Integration : I incorporated an interactive **Google/Wikipedia integration** that allowed users to select a city and access external demographic information. This added contextual depth to the analysis by connecting customer-level findings with broader geographic and demographic information.

◆ THE STORYTELLING

- ➢ The storytelling approach was designed to answer a progressive set of business questions rather than simply display charts. **The story begins with the overall customer landscape, where the distribution of Churned (27%), Joined (6%), and Stayed (67%)** establishes the business context. From there, the analysis moves toward understanding the financial contribution and behavior of these customers.
- ➢ The next layer examines revenue and refund patterns independently to establish a baseline. The analysis then progresses into relationships between **customer tenure, spending, and customer status, helping explain how customer behavior changes throughout the customer lifecycle.**
- ➢ The multivariate analysis adds another level of detail by combining service charges, tenure, customer status, demographic attributes, and customer-level financial measures**. This allows the audience to move beyond simply asking "Who churned?" and instead investigate "What characteristics and behaviors are associated with valuable, retained, or potentially at-risk customers?".**
- ➢ The final stage focuses on high-value customer identification, using revenue and refund rankings to highlight customers who generate strong financial value with relatively low refund activity. Interactive filters allow stakeholders to investigate these patterns across different tenure groups, contract types, and customer statuses. Overall, the storytelling moves from **What is happening? → How are customers behaving? → What relationships exist? → Which customers matter most? → Where should the business focus its attention?**

WHY THIS PROJECT MATTERS

- Customer churn directly affects a telecom company's ability to maintain recurring revenue and maximize customer lifetime value. **A churn rate of 27% represents a meaningful portion of the customer base that is no longer contributing to future revenue**, while the remaining retained customers represent an opportunity for continued growth and relationship development.
- However, churn analysis should not focus solely on the number of customers leaving. Customer value, revenue contribution, refund activity, tenure, service usage, contract structure and **demographics provide important context for understanding the financial impact of customer behavior.**
- This project matters because it connects customer behavior with business value. Instead of treating every customer equally, **the analysis helps identify customers who generate significant revenue, customers who demonstrate higher refund activity and customer segments** where churn or spending patterns may require additional attention.

KEY INSIGHTS

1. Revenue Concentration and High-Value Customers - Univariate Analysis

- The **Total Revenue KPI** indicates approximately **$21.37 million in total revenue**, with an average customer revenue of approximately **$3,034**. Individual customer revenue ranges from **$21 to $11,979**, with the distribution appearing right-skewed.
- This indicates that revenue is not distributed evenly across the customer base. A relatively smaller group of customers contributes disproportionately higher revenue, making these customers particularly important from a profitability and retention perspective.
- **Business Insight:** The business should identify and protect high-value customers because losing a small number of these customers could have a disproportionately larger impact on revenue. At the same time, customers with lower current revenue may represent opportunities for upselling and cross-selling.
- **Business Model Canvas Connection:** Revenue Streams
  Business Impact: Supports high-value customer identification, revenue concentration analysis, and prioritization of revenue expansion opportunities.

2. Low Refund Activity and Customer Relationship Stability - Univariate Analysis

- Total refunds are approximately **$13,819**, with an average refund of approximately **$2** and a maximum refund of **$50**. Relative to the overall revenue generated, refund activity is extremely low.
- This suggests that the business experiences limited revenue leakage through refunds. The low refund levels may indicate effective service delivery, relatively low levels of billing or service dissatisfaction, or restrictive refund policies.
- **Business Insight:** Maintaining low refund activity is financially beneficial because revenue is retained while the business potentially avoids additional costs associated with service dissatisfaction and customer recovery.
- **Business Model Canvas Connection:** Customer Relationships
  Business Impact: Provides an indicator of customer relationship stability and helps monitor potential areas of financial leakage.

3. Tenure and Increasing Monthly Charges - Bivariate Analysis

- The relationship between **tenure and average monthly charges** demonstrates a clear upward trend. Average monthly charges increase from approximately **$49 at month 1 to approximately $80 by month 72**.
- During the mid-tenure period, particularly around **10–30 months**, monthly charges remain relatively stable around the **$60 range**, suggesting that customers may initially settle into mid-tier service plans before increasing their service consumption or adopting higher-value offerings later in their relationship.
- **Business Insight:** Customer longevity appears to be associated with higher monthly revenue. This creates an opportunity to strengthen retention strategies throughout the customer lifecycle and encourage customers to adopt additional services.
- **Business Model Canvas Connection:** Key Activities
  Business Impact: Demonstrates the financial value of customer retention and provides a basis for tenure-based upselling, cross-selling, and loyalty strategies.

4. Customer Status and Lifetime Charges - Bivariate Analysis

- The comparison of **average total charges across customer status** reveals meaningful differences between customer groups. **Stayed customers generate the highest average total charges at approximately $2,788**, indicating that retained customers represent the strongest contribution to cumulative customer value. Churned customers have an average total charge of approximately **$1,531**, which is significant and suggests that customers may leave after generating substantial revenue. Joined customers have the lowest average total charges, which is expected given their shorter customer relationship and limited time to accumulate charges.

- **Business Insight:** The relatively high charges associated with churned customers indicate that churn is not limited to low-value customers. Some customers may leave after establishing meaningful financial value for the business, making retention of these customers particularly important.
- **Business Model Canvas Connection:** Customer Segments
  Business Impact: Supports differentiated strategies for new, retained, and at-risk customers, including stronger onboarding for new customers and targeted retention initiatives for valuable churn-risk segments.

5. Tenure-Based Service Usage - Multivariate Analysis

- The multivariate analysis of **Average Extra Data Charges and Average Long-Distance Charges against Tenure** reveals different patterns in customer service consumption. Long-distance charges demonstrate a **strong upward relationship with tenure ($p < 0.0001$)**, indicating that long-distance usage increases significantly as customers remain with the company longer. Extra data charges also demonstrate a statistically significant but more moderate relationship with tenure **($p = 0.0009$)**. The divergence between the two trends suggests that customers do not increase all services at the same rate. Long-distance usage appears to grow more strongly with tenure, while data-related charges increase at a comparatively slower rate.
- **Business Insight:** Different service categories may require different product and marketing strategies. Long-tenured customers could be targeted with enhanced long-distance offerings, while increasing data consumption provides an opportunity for larger data bundles, premium plans, or personalized upgrades.
- **Business Model Canvas Connection:** Value Proposition
  Business Impact: Supports tenure-based product recommendations, personalized marketing communication, premium-plan opportunities, and long-distance service enhancement.

6. High-Revenue Customers with Minimal Refund Activity - Multivariate Analysis

- The Top 10 customer analysis identifies a small group of customers generating exceptionally high revenue while showing **zero refunds**. These customers demonstrate several characteristics associated with high customer value, including **higher monthly data usage, longer tenure, and unlimited data plans**. This combination is particularly important because these customers contribute significant revenue while generating minimal refund-related revenue leakage.
- **Business Insight:** These customers represent strategic accounts that should receive greater attention from retention and relationship-management programs. Their characteristics can also be used as a reference profile for identifying other customers with the potential to become high-value accounts.

- **Business Model Canvas Connection:** Customer Relationships
  Business Impact: Enables targeted premium-customer retention, identification of high-value prospects, personalized offers, and development of customer segments based on profitable behavioral characteristics.

◆ CHALLENGES & HOW THEY WERE ADDRESSED

1. Moving Beyond Basic Visualization : One of the initial challenges was avoiding the approach of simply creating individual charts and placing them together on a dashboard. I needed to ensure that every visualization answered a meaningful business question.

- **How it was addressed:** I structured the analysis into **univariate, bivariate, and multivariate stages**. This created a logical progression from understanding individual measures to identifying relationships and finally examining multiple customer characteristics simultaneously.
- **What I learned:**I developed a stronger understanding that effective dashboard development begins with the **analytical question**, not the chart type.

2. Making the Univariate Analysis Interactive : I was already familiar with KPI cards and basic aggregation in Tableau, but I wanted the analysis to allow users to examine revenue and refunds from different perspectives without creating multiple separate charts.

- **How it was addressed:** I created the **Revenue & Refund Aggregation parameter**, allowing users to dynamically switch between **Sum, Average, Minimum and Maximum**.
- **What I improved:** This strengthened my understanding of **parameters, calculated logic, and dynamic visualization design**. Instead of treating parameters simply as filters, I learned how they can be used to change the analytical perspective of a visualization.

3. Identifying Meaningful Bivariate Relationships : Another challenge was determining which variable combinations would provide meaningful business insights rather than simply producing visually attractive charts.

- **How it was addressed:**I selected relationships such as Tenure vs. Average Monthly Charges and Customer Status vs. Average Total Charges. I then incorporated **average lines** to establish a benchmark for interpretation.
- **What I learned:** I improved my ability to distinguish between a chart that merely shows a relationship and a visualization that provides a **business comparison or benchmark**.

4. Building Multivariate Analysis Without Losing Clarity : The multivariate analysis was one of the more challenging areas because several variables had to be analyzed simultaneously without making the dashboard difficult to interpret.

- **How it was addressed:** I created a dual-trend visualization comparing **Average Extra Data Charges and Average Long-Distance Charges against Tenure**, using different visual forms and trend lines to distinguish the two service behaviors. I also developed a detailed customer-level table combining revenue, refunds, data usage, age, gender, status, internet type, and unlimited-data information.
- **What I learned:** This improved my understanding of **multivariate visualization, dual-axis analytical design, trend analysis, visual encoding, and information density**.

5. Understanding Customer Value Beyond Churn : Initially, churn could easily be interpreted as simply a percentage of customers leaving. However, the analysis showed that a churned customer can still have generated substantial revenue before leaving.

- **How it was addressed:** I compared customer status with total charges and analyzed revenue and refund behavior at the customer level. This revealed that **churned customers had meaningful average total charges**, demonstrating why the business should evaluate not only how many customers churn, but also which customers churn and how valuable they are.
- **What I learned:** I developed a stronger understanding of **value-based churn analysis** and the importance of connecting customer behavior with financial impact.

6. Identifying High-Value Customers Using Multiple Conditions : Identifying the highest-revenue customers was relatively straightforward. The more challenging requirement was identifying customers who were simultaneously **high revenue and low refund**.

- **How it was addressed:** I created A **Top 10 Limit parameter,** Revenue ranking in descending order, Refund ranking in ascending order, Top 10 filtering and Interactive customer-level analysis. This allowed me to focus on customers who were financially valuable while demonstrating minimal refund activity.
- **What I learned:** I improved my understanding of **ranking, filtering logic, parameter-driven analysis, and customer prioritization**.

- **Tableau Knowledge Strengthened Through the Project :** Although I already had familiarity with Tableau before starting the project, this analysis helped me move toward more advanced and purposeful usage of the platform.

- **Concepts I strengthened :** Dynamic aggregation using **Sum, Average, Minimum and Maximum, Calculated fields and ranking logic, Top-N analysis, Interactive filter, Dashboard actions, KPI design, Average & reference lines, Trend lines and statistical interpretation and Dual-axis visualization.** The major improvement was not simply learning additional Tableau features, but understanding **when and why to use them**. I connected **customer retention with financial value, refund behavior, service consumption, tenure and customer characteristics**. I also combined several analytical perspectives:

Univariate → Bivariate → Multivariate → Customer Ranking → Interactive Exploration → Business Mapping. This approach helped transform a traditional churn dashboard into a broader **customer value and retention analysis**.

- **Overall Learning :** The biggest learning from this project was that advanced analytics is not simply about knowing more Tableau features. It is about understanding **how to combine those features to answer progressively deeper business questions**. I entered the project with knowledge of Tableau's fundamental visualization capabilities, but through this project I strengthened my ability to use **parameters, rankings, dynamic aggregations, trend analysis, statistical interpretation, interactive filters, multivariate visualization, and dashboard storytelling** in a business context. Most importantly, I learned to approach a dashboard as a **decision-support product rather than a collection of charts**—where every visual, interaction, and analytical technique should contribute to understanding the business problem and supporting a potential decision.

STRATEGIC RECOMMENDATIONS

- **1. Implement a Tenure-Based Loyalty & Upgrade Program**
- **Strategic Objective:** Develop a structured loyalty program that recognizes customers as they reach key tenure milestones, particularly **12, 18, and 24 months**. The program can provide targeted incentives such as bonus data, discounted service bundles, upgraded internet tiers, or other benefits based on customer tenure and value.
- **Business Rationale:** The analysis indicates a relationship between tenure and monthly charges, suggesting that customers who remain with the company longer can represent increasing revenue potential. Rather than waiting until customers demonstrate churn behavior, the company can proactively strengthen the relationship at important points in the customer lifecycle.
- **Expected Business Impact:** A **5% improvement in retention among mid-tenure customers** could potentially protect approximately **$500K–$1M in lifetime revenue**, based on the observed increase in monthly charges with tenure.
- **Resource Requirements:** Implementation would require approximately **$50,000 in initial investment**, covering CRM automation, loyalty-tier configuration, reward structures, promotional content, and analytics tracking.
- **Implementation Timeline:**
- **Month 1:** Program design and loyalty-tier development
- **Months 2–3:** Pilot program with selected customer segments
- **By Month 4:** Full-scale rollout and performance monitoring

- **Success Metrics:** 10% reduction in churn among customers with **6–24 months of tenure.** Approximately 8% increase in average monthly charges among participating customers Improvement in retention at 12-, 18- and 24-month milestones

➢ **Business Model Canvas Connection:**
Customer Relationships | Customer Segments | Revenue Streams

2. Expand High-Value Premium Bundles

➢ **Strategic Objective:** Develop premium service bundles combining **Unlimited Data, Fiber Internet, and Long-Distance services**, specifically targeting customers with high data consumption and higher long-distance charges.

➢ **Business Rationale:** The multivariate analysis provides an opportunity to identify customers whose service usage indicates stronger potential demand for premium offerings. Instead of treating these services independently, bundling them can create a more compelling value proposition while increasing revenue per customer.

➢ The strategy should prioritize customers demonstrating **high GB downloads, higher long-distance usage, and existing interest in unlimited-data services**.

➢ **Expected Business Impact:** If approximately **15% of high-usage customers adopt the premium bundle**, the initiative could potentially generate **$2M–$3M in additional annual revenue**, depending on customer adoption and pricing.

➢ **Resource Requirements:** An estimated **$75,000 investment** would support bundle redesign, pricing analysis, promotional campaigns, customer segmentation, and go-to-market activities.

➢ **Implementation Timeline:**

- **Weeks 1–6:** Bundle design, pricing evaluation, and customer segmentation
- **Months 2–3:** Marketing preparation and rollout
- **Month 3 onward:** Adoption monitoring and optimization
- **Success Metrics:** Approximately 20% adoption among targeted high-usage customers. Approximately 7% increase in **Average Revenue Per User (ARPU).** Increased adoption of Unlimited Data and Fiber services. Reduction in churn among customers adopting premium bundles and Incremental revenue generated from bundled services

➢ **Business Model Canvas Connection:**
Value Proposition | Revenue Streams | Cost Structure

3. Introduce a Structured New-Customer Onboarding Journey

- **Strategic Objective:** Create an automated onboarding experience for newly joined customers using **email and SMS communications** to introduce available services, explain data usage, highlight relevant features, and present appropriate upgrade or add-on opportunities.
- **Business Rationale:** New customers represent an important opportunity to establish engagement early in the relationship. A structured onboarding journey can help customers understand the services they have purchased while also introducing relevant products before they become disengaged or develop into potential churn candidates.
- Rather than using a generic communication strategy, onboarding content should be tailored to the customer's service type, usage behavior, and potential upgrade opportunities.
- **Expected Business Impact:** Improving early customer engagement could potentially increase first-year revenue by approximately **$75–$120 per new customer**. At sufficient subscriber scale, this could contribute more than **$1M in additional revenue**.
- **Resource Requirements:** The initiative could be implemented with an estimated investment of **under $40,000**, covering onboarding content, marketing automation, customer education materials, email/SMS workflows, and supporting videos.
- **Implementation Timeline:**
- **Month 1:** Design onboarding journey and communication content
- **Month 2:** Configure automation and conduct testing
- **Month 2 onward:** Deploy and continuously optimize based on customer response
- **Success Metrics:** 15% increase in monthly charges during the first 90 days, 10% increase in add-on adoption, Increased engagement with onboarding communications, Improved early-stage customer retention and Increased adoption of relevant upgrades and services.
- **Business Model Canvas Connection:**
  Channels | Customer Relationships | Revenue Streams

- **Strategic Prioritization**
- Together, these recommendations create a **customer lifecycle-based strategy** rather than relying on a single churn intervention. The onboarding initiative focuses on **new customers**, the loyalty program targets customers progressing through **key tenure milestones**, and premium bundles focus on **high-value and high-usage customers**.This creates three complementary opportunities: **Acquire → Engage → Retain → Expand**
- By connecting customer behavior identified through the analytics with targeted business actions, the organization can move from simply measuring churn to **actively managing customer lifetime value, protecting recurring revenue, and increasing revenue from high-potential customer segments**.

THE CONCLUSION

- The analysis demonstrates that **customer tenure is one of the strongest indicators of revenue, service usage, and long-term customer value**. Across the visualizations, longer-tenure customers consistently demonstrate higher spending and broader service consumption, establishing a clear relationship between customer retention and financial performance.
- The findings suggest that the organization's greatest opportunity for sustainable revenue growth is not solely through acquiring new customers, but through **retaining, nurturing, and expanding relationships with existing customers**. Customers who remain with the company longer generate greater value while also increasing their consumption across multiple service categories. This makes customer retention an important component of long-term profitability and customer lifetime value.
- The analysis also highlights potential warning signals earlier in the customer lifecycle. **Refund activity and fluctuations in monthly charges** can provide indications of dissatisfaction or inconsistent customer experiences. Monitoring these patterns alongside tenure, customer status, and service usage can help the business identify customer segments that may require additional engagement before dissatisfaction develops into churn.
- From a strategic perspective, the results support a shift from a purely acquisition-focused approach toward a more **relationship-oriented customer strategy**. Tailored engagement programs, personalized service offerings, usage-based recommendations, and targeted retention initiatives can be prioritized for customers who demonstrate high current value or strong potential lifetime value.
- The analysis further establishes that customer churn should not be viewed as the result of a single factor. Instead, the findings indicate an interaction between **pricing, tenure, service usage, customer behavior, and customer experience**. Understanding these relationships provides a more complete perspective on why customers may remain, expand their service usage, or eventually leave.
- Future analysis could strengthen this framework by incorporating **predictive churn modeling, customer lifetime value modeling, and deeper demographic and behavioral segmentation**. These approaches could help move the organization from descriptive analysis toward proactive identification of customers who are most likely to churn and the interventions most likely to improve retention.
- Overall, the project brings the individual analyses together into one central business narrative: **Long-term customer relationships drive long-term revenue.** Customers who stay longer not only contribute greater revenue but also tend to expand their service consumption across multiple categories which can play a critical role in **reducing churn, strengthening customer loyalty, increasing lifetime value and improving sustainable profitability**.